Forecasting High-Frequency Bitcoin Volatility: Neural Networks, Market Microstructure and On-Chain Data

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Abstract

We forecast the 30-minute realized volatility of the Bitcoin perpetual future on Binance from five-minute bars built from tick trades, a 20-level order book, funding, open interest and liquidations, together with public on-chain data (mempool, blocks and daily network statistics). Three neural networks (GRU, bidirectional LSTM and a Transformer encoder) are compared with HAR, a linear model and LightGBM, each tuned separately with Optuna and averaged over three seeds. Without on-chain inputs, the self-attention network explains 60.0% of the variance of log realized volatility against 50.8% for HAR, and every non-linear model beats HAR with Diebold–Mariano p-values below 10^{-15}. A linear model with the same 41 features does not beat HAR, so the gain comes from non-linearity rather than from the extra information. An ablation over three feature sets shows that adding 13 intraday on-chain variables raises the out-of-sample squared error of every model by 0.7–7.9% ($p < 10^{-8}$), while daily on-chain series have no measurable effect. Public on-chain data do not help, and actually hurt, short-horizon volatility forecasts.

Keywords: realized volatility; Bitcoin; deep learning; Transformer; HAR; market microstructure; on-chain data.

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1 Introduction

Bitcoin returns over a few minutes are close to unpredictable, but their volatility is not. In earlier work on the same data, no model beat the zero forecast of 30-minute returns; a weak but significant directional signal did not survive transaction costs. Volatility, by contrast, is strongly persistent (Andersen and Bollerslev, 1998), which makes it the natural target on which to test whether complex models add value.

This paper asks three questions:

1. Do recurrent and attention-based networks beat HAR (Corsi, 2009), the standard benchmark for realized volatility, and a gradient-boosted tree model (Ke et al., 2017) that uses the same features?

2. Do public on-chain data (mempool, blocks, daily network statistics) improve the forecast beyond microstructure and derivatives data?

3. If on-chain data matter, is it the intraday block or the daily block?

The contributions are threefold. First, a look-ahead-free pipeline that aligns minute-level market data with on-chain data at the moment each value becomes known. Second, a comparison of seven forecasts with separate hyperparameter tuning and multi-seed averaging. Third, a feature-group ablation evaluated with Diebold–Mariano tests that are robust to the overlap of the forecast targets.

The main findings are that non-linear models beat HAR by about 0.08 in R^2 and by 25–30% in QLIKE, that networks and LightGBM perform almost equally well, and that on-chain data consistently worsen the forecasts.

2 Data

The sample covers two years, January 2024 to December 2025, on the BTC-USDT perpetual future of Binance Futures. All sources are aggregated to one minute and then to five-minute bars, about 210,000 bars in total. Table 1 lists the sources.

Table 1. Data sources.

Source	Table	Derived quantities	Frequency
Crypto Lake, Binance Futures	trades	OHLC, VWAP, buy and sell volume, number of trades, largest trade, tick realized variance	tick
Crypto Lake, Binance Futures	book_1m	spread, microprice, imbalance at 1/5/10/20 levels, depth, book slope	1-min snapshot
Crypto Lake, Binance Futures	funding, open interest, liquidations	funding rate, change in open interest, long and short liquidations	event
Blockchain.com	charts API	mempool size in bytes, pending transactions, transactions per second	15 min
mempool.space	blocks	block interval, fullness, fee rate, total fees, output value, UTXO change (111,155 blocks)	per block
Blockchain.com	charts API	transactions, fees in USD, hash rate, unique addresses, estimated volume, block size	daily

Look-ahead-free alignment. Each on-chain value enters a bar only from the moment it is known, through an as-of merge on the bar's closing time. The 15-minute mempool series is an average over the quarter hour, so it becomes known at the end of that quarter hour; values older

than one hour are discarded. Block timestamps are set by miners and can be inaccurate, so a block is treated as known 20 minutes after the running maximum of previous timestamps. The daily value of day D is known only from 00:00 of day $D + 1$.

Missing data. Crypto Lake lacks about 7 days of order book out of 731 and 8 hours on 13 November 2024. A value is carried forward for at most one hour; beyond that it is set to the training median, and a flag `book_ok` tells the model whether the book is observed.

3 Features and target

Each bar carries up to 60 features in seven groups (Table 2). Every feature uses only information available at the close of the bar.

Table 2. Feature groups. The code is used to name the feature sets of the ablation.

Group	Code	N.	Features
Price	P	10	returns over 1, 3, 6, 12, 48 bars; volatility over 12 and 48 bars and their ratio; high–low range in bps; log tick realized volatility
Flow	F	8	order-flow imbalance over 1, 6, 24 bars; volume and trade count relative to the daily mean; average trade size; share of the largest trade; distance from VWAP
Book	B	12	spread at bar end and on average; microprice deviation; imbalance L1/L5/L10/L20 and change in L5; depth L5; USD imbalance at L20; bid and ask slope; `book_ok`
Derivatives	D	6	funding rate; change in open interest over 1 hour and 1 day; long, short and net liquidations
Intraday on-chain	Oi	13	mempool size relative to the daily mean, its 1-hour change, pending transactions, arrival rate; means over the last 6 blocks of interval, fullness, fee rate, fees, output value, UTXO change, transactions; blocks in the last hour; minutes since the last block
Daily on-chain	Og	6	30-day z-scores of log transactions, fees in USD, hash rate, unique addresses, estimated volume, block size
Calendar	C	4	sine and cosine of time of day and day of week

Target. The target is the log realized volatility over the next $H = 6$ bars (30 minutes), computed on the order-book mid-price m_t to remove bid–ask bounce:

$$y_t = \tfrac{1}{2} \log \sum_{j=1}^{H} (\log m_{t+j} - \log m_{t+j-1})^2 . \tag{1}$$

The logarithm makes the distribution nearly symmetric and stabilizes its variance; the loss is the squared error on the standardized target.

Preprocessing. Features are scaled robustly with statistics estimated on the training block only (median, interquartile range, clipping at ± 10); remaining missing values are set to the median. The target is standardized with the training mean and standard deviation.

4 Models

Three neural networks are compared with three benchmarks of increasing complexity (Figure 1). The networks read a window of the last L bars ($L = 24$ or 48, i.e. 2 or 4 hours); the linear model

and LightGBM use the last bar only.

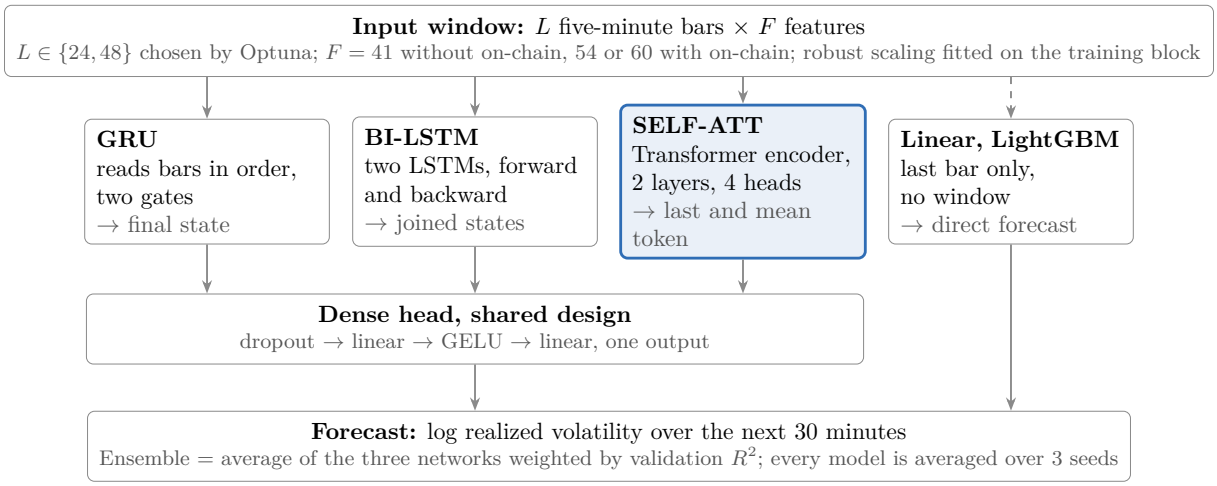


Figure 1. Model architecture. The highlighted network is the best model in the test set. HAR is estimated separately on past realized volatility only.

4.1 Benchmarks

HAR. The heterogeneous autoregressive model of Corsi (2009) regresses future log volatility on three averages of past realized volatility over different horizons, estimated by OLS:

$$y_t = \beta_0 + \beta_1 \log RV_t^{(30\,\mathrm{min})} + \beta_2 \log RV_t^{(1\,\mathrm{day})} + \beta_3 \log RV_t^{(1\,\mathrm{week})} + \varepsilon_t. \tag{2}$$

It captures the persistence of volatility with four parameters and is the standard benchmark in the literature.

Linear. A linear combination of all features of the last bar, trained with weight decay, i.e. a ridge regression. It measures the value of the additional information without non-linearity.

LightGBM. Gradient-boosted regression trees (Ke et al., 2017) on the same last-bar features; each tree fits the residual of the previous ones and early stopping on the validation block selects the number of trees. It is the strongest non-linear benchmark on tabular data.

4.2 Neural networks

GRU. A gated recurrent unit (Cho et al., 2014) reads the window one bar at a time and updates a hidden state h_t. An update gate z_t decides how much of the old state to keep and a reset gate r_t how much to use when computing the candidate:

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \qquad r_t = \sigma(W_r x_t + U_r h_{t-1}), \tag{3}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tanh\big(W x_t + U(r_t \odot h_{t-1})\big). \tag{4}$$

The final state is passed to the dense head.

BI-LSTM. Two long short-term memory networks (Hochreiter and Schmidhuber, 1997) read the window forward and backward, and their final states are concatenated. The LSTM has a separate memory cell and three gates (input, forget, output). Reading backward uses no future data: the window still ends at time t.

SELF-ATT. A two-layer Transformer encoder (Vaswani et al., 2017) with four heads and pre-layer normalization. Each bar is linearly projected and added to a learned positional embedding; each layer computes the attention of every bar on all the others,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \tag{5}$$

Unlike a recurrent network, the model can directly relate two distant bars of the window, such as a volume spike two hours earlier. The head receives the last token concatenated with the mean of all tokens.

Ensemble. The average of the three networks with weights proportional to their validation R^2 (about one third each in all runs).

4.3 Training

Networks are trained with AdamW (Loshchilov and Hutter, 2019), batches of 512 samples and every second training sample (stride 2), which reduces the overlap of input windows. Regularization combines dropout, weight decay, Gaussian noise on the inputs during training and gradient clipping at 1. The learning rate is halved when the validation loss stops improving, and early stopping with a patience of 6 epochs (maximum 40) keeps the best weights. Every model is trained with three seeds (42, 43, 44) and the forecasts are averaged.

5 Experimental design



Figure 2. Chronological split. The training block fits the model weights, the feature scaling and the target standardization; the validation block drives tuning, early stopping, ensemble weights and feature importance; the test block is used only for the final evaluation. Between blocks, samples whose target crosses the boundary are purged and an embargo of 6 bars is applied.

Split, purging and embargo. The sample is divided into three consecutive blocks (Figure 2). A training sample is dropped if its target, which extends to $t + H$, crosses into the next block (purging), and an additional embargo of $H = 6$ bars separates the blocks (López de Prado, 2018). The training block has 146,925 samples (2 January 2024 – 26 May 2025), the validation block 31,524 (27 May – 13 September 2025) and the test block 31,524 (13 September – 31 December 2025).

Tuning. Each network and LightGBM receive 20 Optuna trials (Akiba et al., 2019) with a TPE sampler and a median pruner (10 epochs per trial), i.e. 80 configurations per run. The search space favours small, heavily regularized networks: 8–48 units, dropout 0.2–0.6, learning rate $3 \cdot 10^{-5}$–$5 \cdot 10^{-4}$, weight decay 10^{-4}–10^{-1}, input noise 0–0.2, window of 24, 48 or 96 bars. The criterion is the validation loss, and every feature set is tuned from scratch with the same budget.

Metrics. On the test block we report (i) R^2 relative to the training mean, the only mean known at forecast time; (ii) the QLIKE loss, robust to noise in the volatility proxy (Patton, 2011); (iii) Spearman's rank correlation ρ; and (iv) the Diebold–Mariano statistic (Diebold and Mariano, 1995) on the squared-error differential, with a Newey–West variance (Newey and West, 1987) at $H - 1 = 5$ lags because the targets of neighbouring bars overlap:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{Var}}_{NW}(\bar{d})}}, \quad d_t = e_{1,t}^2 - e_{2,t}^2, \qquad \mathrm{QLIKE} = \frac{1}{T}\sum_t \left(\frac{\sigma_t^2}{\hat{\sigma}_t^2} - \log \frac{\sigma_t^2}{\hat{\sigma}_t^2} - 1 \right). \tag{6}$$

Ablation. Three runs are identical except for the feature groups (Table 3). Run A was executed twice and gave metrics identical to the last digit: given seeds and hyperparameters, the pipeline is deterministic.

Table 3. Feature sets of the ablation.

Run	Groups	Code	Features
A, full	all	PFBDOiOgC	60
C, intraday on-chain only	all but Og	PFBDOiC	54
B, no on-chain	all but Oi and Og	PFBDC	41

6 Results

6.1 Model comparison

Table 4 reports test metrics for run B, the best feature set. SELF-ATT has the highest R^2 and rank correlation, and every network and LightGBM beat HAR with $p < 10^{-15}$. Relative to the test mean, all R^2 values are about 0.035 lower, but the ranking is unchanged.

Table 4. Test results, run B (no on-chain), 31,524 forecasts. DM is one-sided against HAR: positive means the model beats HAR.

Model	R^2 (train mean)	R^2 (test mean)	QLIKE	ρ	DM vs HAR	p
SELF-ATT	**0.600**	**0.566**	0.562	**0.749**	19.7	$< 10^{-15}$
Ensemble	0.594	0.560	0.561	0.749	19.8	$< 10^{-15}$
LightGBM	0.588	0.553	**0.553**	0.745	16.9	$< 10^{-15}$
BI-LSTM	0.586	0.550	0.582	0.742	17.6	$< 10^{-15}$
GRU	0.582	0.546	0.564	0.743	16.9	$< 10^{-15}$
Linear	0.509	0.467	0.623	0.710	0.15	0.44
HAR	0.508	0.466	0.773	0.676	–	–

Three results stand out.

- **Non-linearity is what matters.** The linear model with the same 41 features does not beat HAR ($p = 0.44$). All non-linear models gain about 0.08 in R^2 and reduce QLIKE by 25–30%.
- **Networks and LightGBM are on par.** SELF-ATT has the highest R^2 and correlation; LightGBM has the lowest QLIKE, i.e. it errs less when it underpredicts volatility.
- **The time window adds little.** LightGBM sees only the last bar and ends 0.012 below SELF-ATT, which sees 48: almost all the information is already in the features of the current bar.

Figure 3 shows two test days. HAR reacts to spikes with a lag because it depends only on averages of past volatility; SELF-ATT anticipates part of the move through the range, volume and trade count of the current bar. Figure 4 shows that its forecasts are more tightly concentrated around the diagonal.

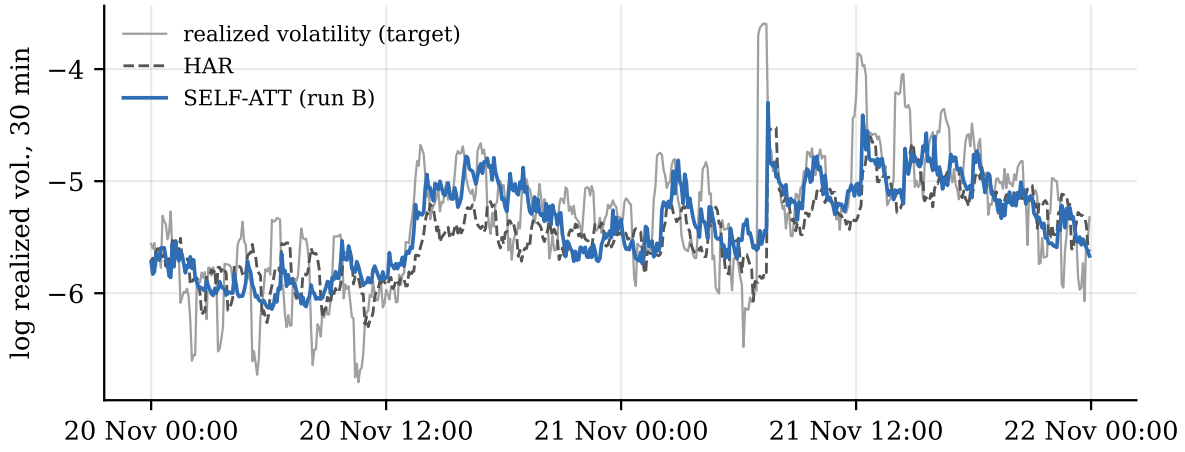


Figure 3. Realized volatility, HAR and SELF-ATT forecasts on 20–21 November 2025 (test block, run B).

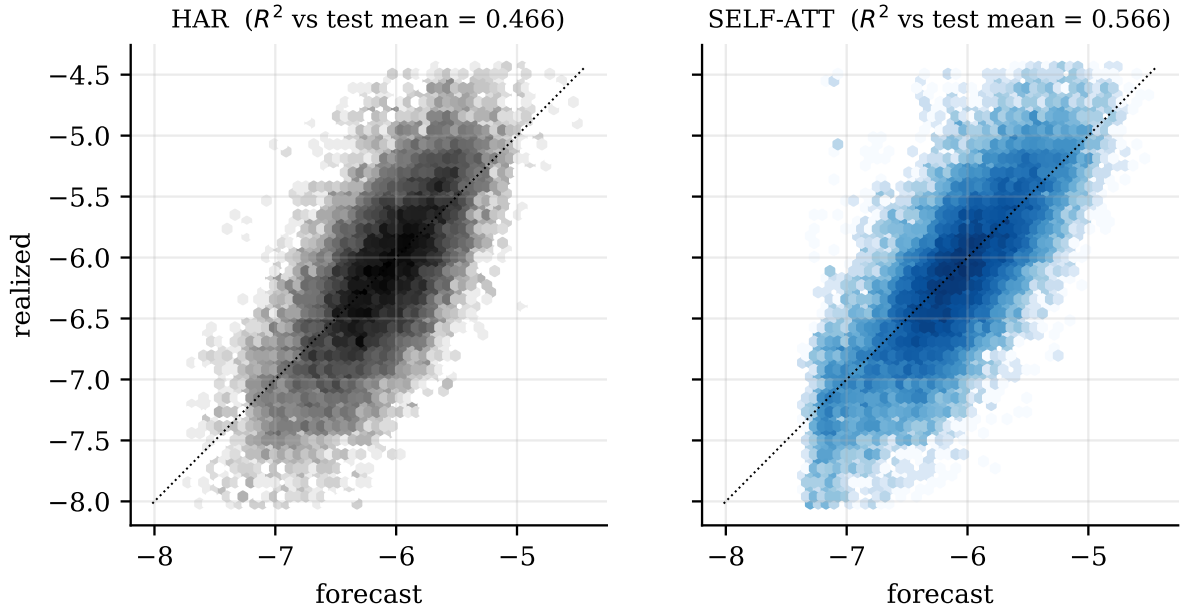


Figure 4. Forecast against realized log volatility on the test block, run B. Colour intensity is the log count of observations per cell; the dotted line is the diagonal.

6.2 Stability over time and across regimes

The advantage does not come from a few episodes: SELF-ATT has a higher daily R^2 than HAR on 103 of the 110 test days (Figure 5). The gain relative to HAR is largest in extreme regimes (Figure 6). In the most volatile decile SELF-ATT reduces the squared error by 30% and LightGBM by 33%; in the calmest decile SELF-ATT reduces it by 17%; in the middle deciles the gain is 6–11%.

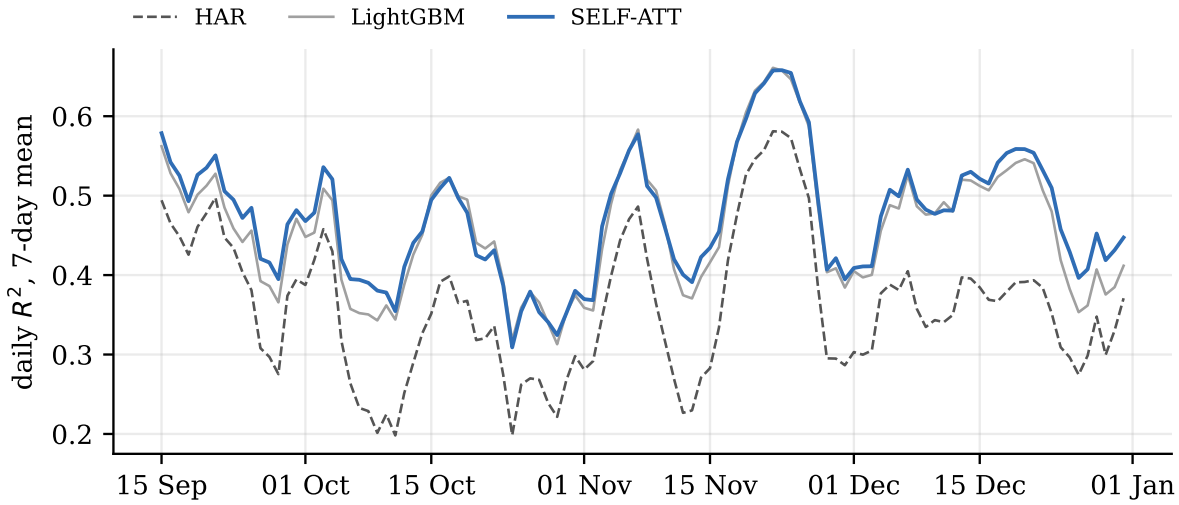


Figure 5. Daily R^2 on the test block, 7-day moving average, run B.

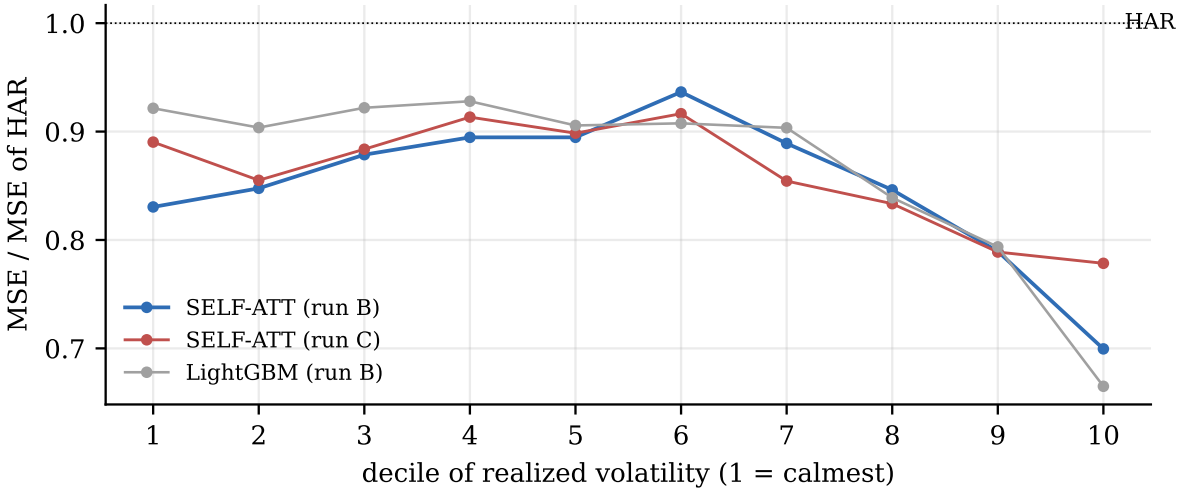


Figure 6. Test mean squared error by decile of realized volatility, relative to HAR. Values below 1 mean the model beats HAR in that decile.

6.3 Ablation: the effect of on-chain data

Adding the 13 intraday on-chain variables (run C) increases the test loss of every model relative to run B; adding the 6 daily series as well (run A) changes almost nothing relative to C (Figure 7 and Table 5).

Table 5. Diebold–Mariano tests between feature sets (two-sided, Newey–West with 5 lags). For C vs B a negative statistic means C is worse.

	GRU	BI-LSTM	SELF-ATT	LightGBM	Linear	Ensemble
C vs B, DM	-6.5	-14.0	-7.6	-6.4	-5.8	-12.7
C vs B, p	$8 \cdot 10^{-11}$	$< 10^{-15}$	$4 \cdot 10^{-14}$	$2 \cdot 10^{-10}$	$8 \cdot 10^{-9}$	$< 10^{-15}$
C vs A, Δ MSE	-0.6%	-0.2%	$+0.7\%$	-0.5%	$+0.4\%$	$+0.2\%$
C vs A, p	0.04	0.47	$6 \cdot 10^{-5}$	$2 \cdot 10^{-5}$	10^{-6}	0.28

The differences between A and C are below 1% and change sign across models: the daily series neither add nor remove information, and the damage comes entirely from the intraday

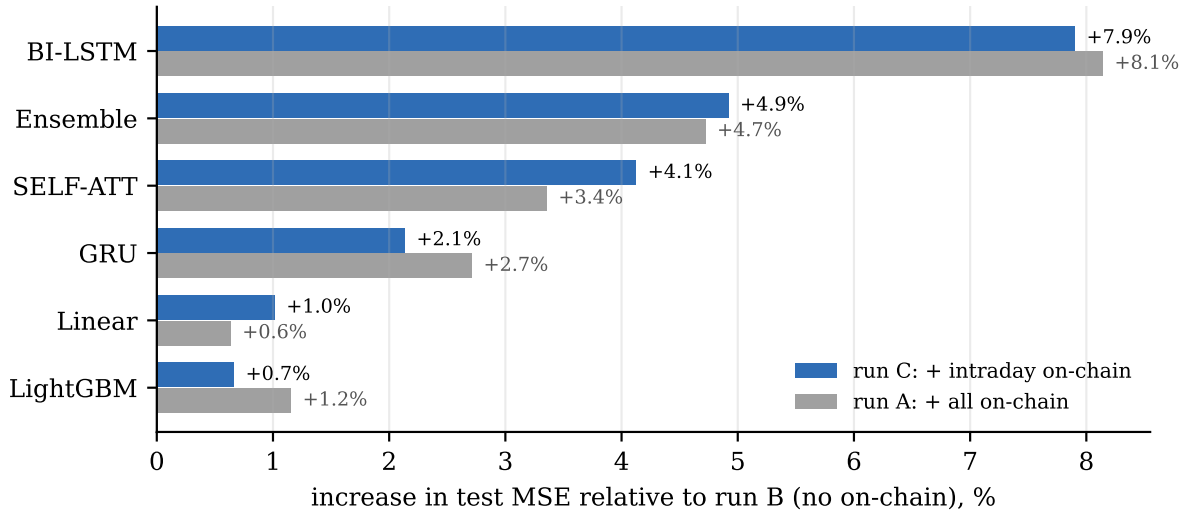


Figure 7. Increase in test mean squared error of runs A and C relative to run B. All differences against B are significant (Diebold–Mariano $p \leq 6 \cdot 10^{-4}$).

block. Figure 8 shows that the deterioration is not driven by a single episode: the cumulative squared-error difference grows throughout the four test months, with jumps on days of high volatility. Figure 6 shows where: with on-chain inputs, SELF-ATT loses mostly in the extreme deciles (from 0.70 to 0.78 of HAR's error in the most volatile one).

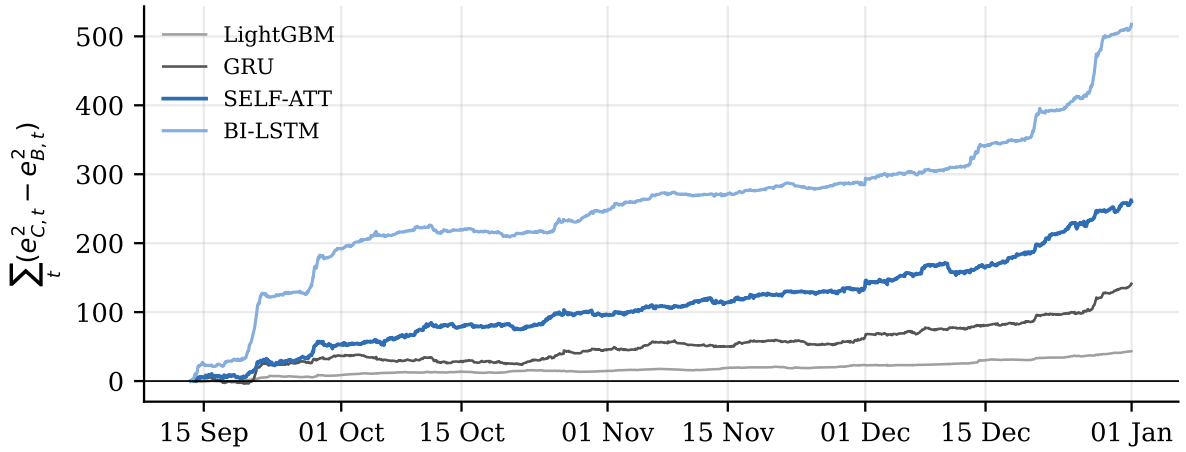


Figure 8. Cumulative difference in squared error between run C and run B on the test block. An increasing line means run C (with intraday on-chain data) is accumulating more error.

6.4 Feature importance

Permutation importance is computed on the best network in validation: one feature at a time is shuffled and the percentage increase in validation loss is recorded (Figure 9). The high–low range of the bar dominates (34–36%), followed by recent volatility, trading activity and calendar effects. In run C no on-chain variable exceeds 1%. The most harmful is the average fullness of the last six blocks (−0.36%): shuffling it improves the model. The relative mempool size had an importance of −0.27% in run A and +0.33% in run C; a sign that flips between runs points to an unstable relationship.

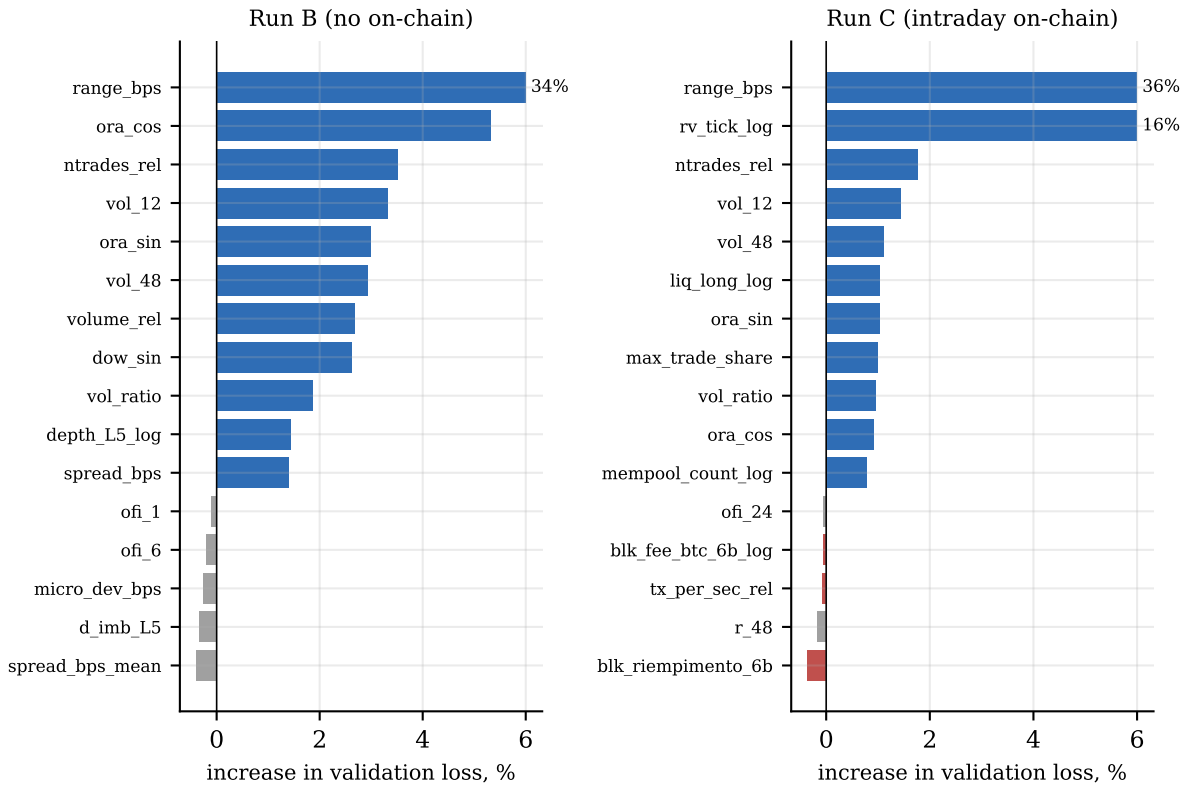


Figure 9. Permutation importance in runs B and C: the five least and eleven most important features. Bars are capped at 6% (values written beside them); red marks on-chain variables with negative importance.

6.5 Training curves

The validation loss stabilizes within 5–10 epochs while the training loss keeps decreasing (Figure 10). Early stopping ends GRU and BI-LSTM after about 14 epochs and SELF-ATT after 23. The gap between training and validation loss is small (0.05–0.07 in standardized units), which indicates that regularization is effective.

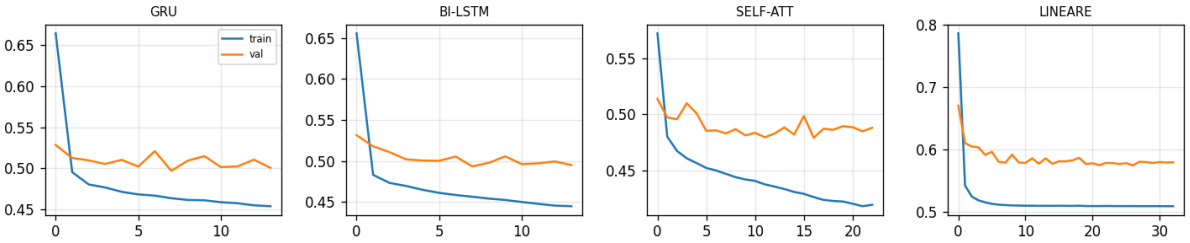


Figure 10. Training and validation loss per epoch, run B, first seed ("LINEARE" is the linear benchmark).

7 Discussion

The explanation most consistent with the evidence is overfitting to unstable relationships, not a lack of signal in the models. Three observations point the same way. First, on-chain variables have importance close to zero and sometimes negative, with a sign that changes between runs: the model finds relationships in the training block that no longer hold in validation and test. Second, the damage grows with model flexibility: +7.9% for BI-LSTM, +4.1% for SELF-ATT and +0.7% for LightGBM, which selects its own features in each split. This is the typical pattern

of overfitting. Third, the damage is concentrated in extreme regimes, where the model should rely on the market but instead weighs variables that, in the training sample, were associated with different episodes.

There is also an economic reason. At a 30-minute horizon, Bitcoin volatility is driven by derivatives trading, which takes place almost entirely off-chain. Mempool and block statistics measure the demand for block space and are only indirectly linked to short-term price moves. Blocks arrive on average every ten minutes and with a publication delay, so any useful information is likely to be already in the price.

Robustness. Hyperparameters are tuned from scratch for every feature set, so the deterioration is not due to settings chosen for another configuration. Forecasts are averaged over three seeds, and a repeated run gives identical metrics. Diebold–Mariano tests use HAC standard errors for the overlapping targets. The result holds for all five trained models, including the linear model and LightGBM, and for the ensemble.

Limitations. The evaluation uses a single time split: the test block covers four months (September–December 2025) of one market regime, and a walk-forward evaluation over several windows would make the result more general. Only one horizon (30 minutes) and one instrument are studied; on-chain data may matter more at horizons of days. The on-chain features are public and free; proprietary metrics such as exchange inflows and outflows were not tested. The tuning budget is 20 trials per model; with more trials the networks might learn to ignore uninformative inputs. Finally, the forecasts are not translated into a trading or risk-management application, so their economic value remains to be measured.

8 Conclusion

At a 30-minute horizon, Bitcoin volatility is predictable, and non-linear models predict it much better than HAR. The self-attention network explains 60.0% of the variance of log volatility against 50.8% for HAR; LightGBM is almost on par and has the best QLIKE. The gain comes from non-linearity: a linear model with the same features does not beat HAR.

Public on-chain data do not help. Mempool and block statistics significantly worsen the forecasts of every model, and daily network series have no effect. The best feature set combines microstructure and derivatives data, with 41 variables. Natural extensions are a walk-forward evaluation, longer horizons where slower on-chain information may matter, exchange-flow data instead of mempool and block statistics, and the use of the forecasts in volatility targeting or market making to measure their economic value.

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A Hyperparameters and reproducibility

Table 6 reports the hyperparameters selected by Optuna in run B. For LightGBM the selected values are a learning rate of 0.029, 13 leaves, at least 170 observations per leaf, 79% of the features per tree and an L2 penalty of 0.62, with a validation loss of 0.488.

Table 6. Selected hyperparameters of the networks, run B.

Model	Units	Dropout	Learning rate	Weight decay	Input noise	Window (bars)	Val. loss
SELF-ATT	48	0.21	$3.0 \cdot 10^{-4}$	$1.4 \cdot 10^{-3}$	0.12	48	0.481
BI-LSTM	48	0.44	$2.9 \cdot 10^{-4}$	$3.9 \cdot 10^{-2}$	0.14	24	0.493
GRU	32	0.34	$4.9 \cdot 10^{-4}$	$7.1 \cdot 10^{-4}$	0.01	24	0.497

Code and data. `scarica_cryptolake.py` downloads the Crypto Lake tables and aggregates them to one minute; `scarica_onchain.py` downloads mempool, block and daily series; the notebook `modelli_btc_hf.ipynb` runs the whole pipeline, and the three runs differ only in the `GRUPPI_FEATURE` setting. Figures 3–9 are computed by `figures.py` from the saved test predictions; Figure 10 is produced by the notebook.